

Biology of the electric eel, *Electrophorus electricus*, Linnaeus, 1766 (Gymnotiformes: Gymnotidae) on the floodplain of the Curiaú River, eastern Amazonia

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Received: 9 April 2015 / Accepted: 6 November 2015 / Published online: 17 November 2015
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Abstract The evaluation of the growth patterns and reproductive strategies of fish species are vitally important for the understanding of their biology and the management of stocks. The present study focused on the Amazonian electric eel (*Electrophorus electricus*), which is capable of producing an electrical discharge of up to 800 V. Specimens were collected on a monthly schedule from a floodplain in the eastern Amazon basin. The gonads of these specimens were examined, and the sex ratio, growth parameters, population structure, body size at first gonadal maturation, and the gonadosomatic index were determined. A balanced sex ratio was found. Males were larger than females, and both sexes presented isometric growth, which is unusual in species with an elongated anatomy. This isometry may be related to the reduction of the coelomic cavity, and its position near the

head. The spawning period coincided with the start of the rainy season, and continued until high water, with the variation in gonadal development following the fluctuations in precipitation and river water levels. The asymptotic body length in both sexes was relatively large, and was inversely related to the growth coefficient (k), with a slower growth rate being recorded in the males. Mortality rates were relatively low in comparison with most species of tropical fish. The larger size of the male may be related to their role in parental care, and sexual selection on the part of the females. These findings may be important for the management of wild stocks, as well as captive rearing.

Keywords Electric eel · Growth · Mortality · Weight-length ratio · Gonadosomatic index · Amapá

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Introduction

The Amazon basin contains the greatest diversity of freshwater fishes in the world (Lowe-McConnell 1999), although relatively little is known of the biology of most species, except for those of economic importance. One of the least well-known species is the electric eel, *Electrophorus electricus*, which is abundant in Amazonian ecosystems, where it is feared by the local population due to its electrical discharge, as well as its mythical status, in addition to being used as a medicinal component of the treatment of a number of

different disorders. *Electrophorus electricus* is endemic to the Neotropical region, with a total body length of up to 2.5 m, and the capacity to produce electrical charges of up to 650 V (Campos-da-Paz 2003). The body of *E. electricus* is anguilliform, and there is a long anal fin with more than 250 soft rays (Fig. 1). De Santana et al. (2013) recently discovered vestiges of the bones of a true tail, contradicting the traditional view that the species has always lacked a caudal fin (Mago-Leccia 1994; Albert 2001; Campos-da-Paz 2003). Electric eels are nocturnal (Assunção and Schwassmann 1995) and are obligate air breathers, for which they use the highly vascularized oral region (Johansen 1970). This allows the electric eel to inhabit hypoxic backwaters in the Amazon and Orinoco basins (Mago-Leccia 1994; Crampton 1996; Campos-da-Paz 2003). While electric eels are classified generally as piscivores (Mago-Leccia 1994), it is likely that they are generalist carnivores, feeding on fish, small tetrapods, crustaceans, and terrestrial and aquatic insects (Planquette et al. 1996; Mérona and Rankin-de-Mérona 2004). They may also occasionally consume the fruit of the açai palm, *Euterpe oleracea* Mart. (Nakashima 1941; Goulding 1980).

The anatomy and electrophysiology of the electric organ of *E. electricus* have been well understood since the nineteenth century (Sachs and Fritsch 1881), although there are few data on reproductive patterns in this species, except for the studies of Assunção and Schwassmann (1995) and Sá-Oliveira and Mendes-Jr (2012). The males build nests and provide parental

care, and are larger and more numerous than the females (Assunção and Schwassmann 1995). Fertility is 1730–3063 oocytes (Sa-Oliveira and Mendes Jr-2012). There is no consensus on spawning patterns. Whereas Assunção and Schwassmann (1995) concluded that the species has parceled spawning during the dry season, Sá-Oliveira and Mendes-Jr (2012) concluded that it was of the total type and occurred at the beginning of the dry season (see also Sachs and Fritsch 1881).

The powerful electrical discharge of *E. electricus* fuels fears among local human populations, and leads to the indiscriminant killing of these eels in an attempt to minimize the risk of accidents. The impacts caused by these attempts to control *E. electricus* populations are poorly understood, and studies such as the present one will be essential for the development of effective measures for the management of populations. The present study provides data on a number of morphological, physiological and reproductive and growth parameters of the *E. electricus* population from the floodplain of the Curiaú River in Macapá, in the Brazilian state of Amapá, located in the northeastern Amazon basin.

Materials and methods

Study area

The Rio Curiaú Environmental Protection Area (APA do Curiaú) is located in the municipality of Macapá in

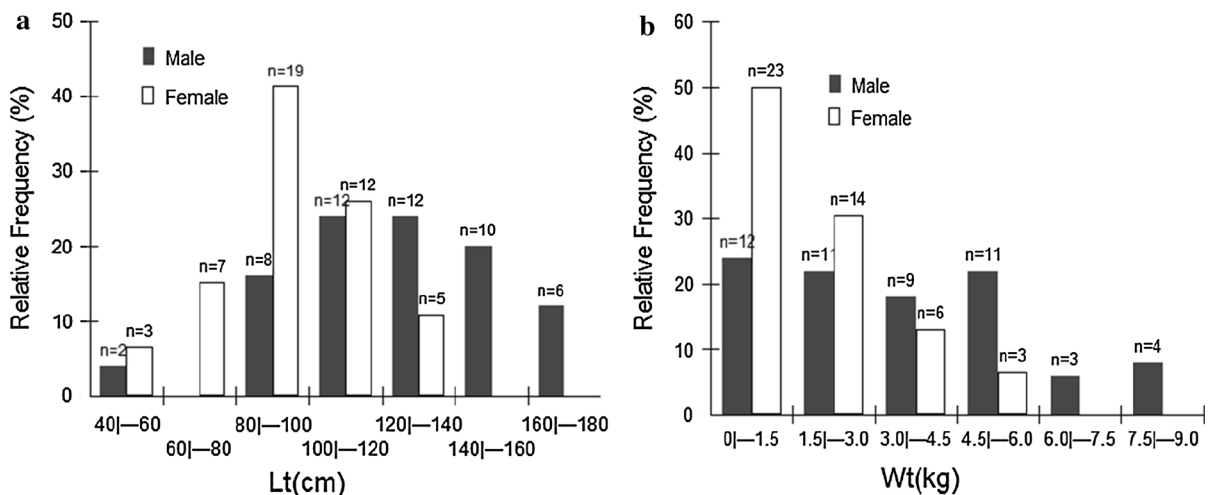


Fig. 1 Distribution of **a** total length (Lt) and **b** total weight (Wt) values in the male and female *E. electricus* specimens

the Brazilian state of Amapá, in northeastern Amazonia. This sustainable use protected area encompasses a total of 23,000 ha, extending to latitudes 00°14'58"N in the south and 00°14'17"N in the north, and longitudes 50°56'54"W in the east, and 51°07'46"W, in the west (Fig. 2). The area is drained by the basin of the Curiaú River, which includes areas of Amazonian savanna, *terra firme* and swamp forests, and seasonally flooded grasslands, as well as temporary and permanent lentic bodies of water (Chellappa et al. 2005). This diversity of environments supports a rich and abundant fauna within the APA do Curiaú.

Specimen collection

Specimens of *E. electricus* were collected every 2 months between March, 2005, and February, 2006, using a variety of fishing equipment, including gillnets set overnight, shrimping nets, hand-nets, cast nets, and hand lines. The different types of fishing gear were used due to the difficulty and dangers of capturing this nocturnal species. The shrimp net (0.5 cm mesh) was used to capture individuals of all sizes, juveniles and adults, thus avoiding sampling bias. All captured specimens were taken to the Limnology and Oceanography Laboratory at the Federal University of Amapá (LLO-UNIFAP), where they were conserved in ice until analysis. Data on precipitation for the study period were obtained from the meteorological station of the Amapá State Research Institute (IEPA) in Macapá.

Each specimen was measured [total length (L_T) in cm] using a tape measure and calipers, and weighed (Wt in kg) using a Filizola BP-15 precision balance (precision of 5 g). The sex and gonadal maturation of

each specimen was determined through a macroscopic visual analysis of the gonads, based on the protocol of Vazzoler (1996). The gonads were weighed on a BEL 500c balance (precision of 0.01 g). Voucher specimens were deposited in the Biological Collection of the National Amazonian Research Institute (INPA) in Manaus, Brazil, under catalog number INPA 41112 and 41124.

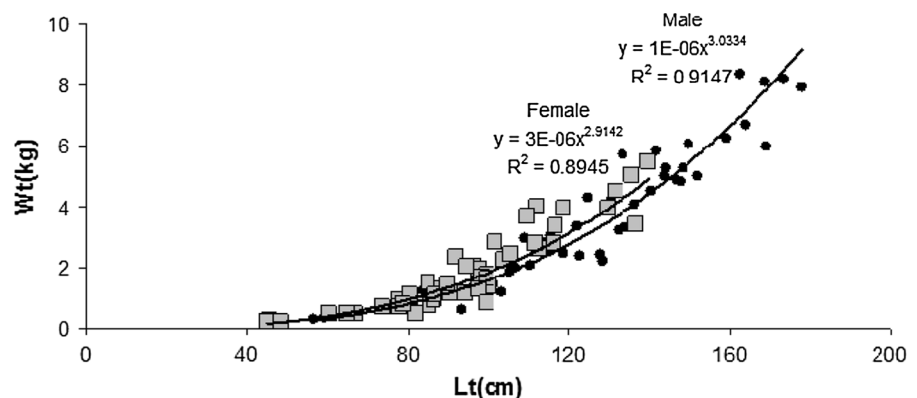
Data analysis

Possible deviations in the sex ratio were tested using Chi square, and sexual dimorphism was evaluated using a one-way Analysis of Variance (ANOVA) for body length and the nonparametric Kruskal–Wallis analysis of variance for body weight. In all cases, $\alpha = 0.05$.

The weight-length relationship was analyzed separately for males and females, considering the equation $Wt = a \times L_T^b$. The data were subsequently log-transformed for the analysis of growth patterns, based on the b coefficient, where $b = 3$ is considered to represent isometric growth, $b < 3$ a negative allometric growth pattern, and $b > 3$, a positive allometric pattern (Zar 1996).

Growth parameters were based on the Von Bertalanffy equation, $L_t = L_\infty \times (1 - e^{-k(t-t_0)})$ (Sparre and Venema 1998), where L_t = the total length of the fish (cm) at age t , L_∞ = the asymptotic length (cm), k = growth rate (year^{-1}), and t_0 = the nominal age at which the size of the fish is considered to be zero. The constant k and L_∞ and W_∞ were calculated in the ELEFAN (Electronic Lengths-Frequency Analysis) routine (Pauly and David 1981) available in the FISAT (FAO-ICLARM Stock Assessment Tools) program

Fig. 2 Weight–Length relationship for the **a** female and **b** male *Electrophorus electricus* specimens from the APA do Curiaú, Amapá, Brazil



(Gayanilo et al. 1996; Gayanilo and Pauly 1997), which is based on the modal displacement of the temporal sequences of body length (Castro et al. 2002). Longevity ($A_{0.95}$) was calculated as in Taylor (1958), by $A_{0.95} = t_0 + (2.996/k)$, and natural mortality (M) was estimated using Pauly's (1980) formula: $\ln MP = -0.0152 - 0.279 \times \ln L_{\infty} + 0.6543 \times \ln k + 0.463 \times \ln T$, where T is the mean surface temperature, in this case, 28 °C.

Mean length at first sexual maturation (L_{50}) was estimated in both sexes based on the total length (L_T , in cm) of the individuals that had already reached first gonadal maturity, following Vazzoler (1996). The Gonadosomatic Index (GSI) was also calculated for males and females, as described by Wootton et al. (1978) and Vazzoler (1996).

Mean age at first gonadal maturation (t_{50}) was estimated using von Bertalanffy's inverse model (Sparre and Venema 1998): $t_{50} = t_0 - (1/K) \times \ln(1 - L_{50}/L_{\infty})$, where L_{∞} = the maximum theoretical body length, t_0 = theoretical age at body length zero, K = the growth constant, and L_{50} = mean body length at first gonadal maturation, at age t_{50} .

The mean GSI values recorded every 2 months were used to define the breeding season (Vazzoler 1996), based on a one-way ANOVA ($\alpha = 0.05$), with bimesters being classified as flood (January/February), high water (Mar/April, May/June), ebb (July/August, September/October), and low water (November/December) seasons. To investigate seasonal influences on the GSI and weight of the *E. electricus* specimens, an Analysis of Covariance (ANCOVA) was run using the GSI as the response variable, which was analyzed in relation to seasonal variation and body weight. For this analysis, the GSI and body weight values were transformed by their natural logarithms ($\ln$). To determine whether the GSI was affected by rainfall, the values were compared with the precipitation recorded during the previous month using a Spearman correlation coefficient.

Results

A total of 96 *E. electricus* specimens were collected during the study period, of which 46 were female, and 50 male. The sex ratio did not deviate significantly, considering either the rainy (1.04 F: 1 M, $\chi^2 = 0.96$; $p > 0.05$) or dry (0.82 F: 1 M, $\chi^2 = 0.87$; $p > 0.05$)

seasons, or the study period as a whole (0.92 F:1 M, $\chi^2 = 0.94$; $p > 0.05$).

Mean total length (Fig. 1) in the females was 94.3 ± 23.6 cm (range 45–140 cm), considerably lower than in the males which measured 124.2 ± 28.3 cm, on average (range 56.0–177.7 cm). The mean weight (Fig. 4) of the females (1.92 ± 1.36 kg; range 0.21–5.49 kg) was also much lower than that of the males (3.62 ± 2.0 kg; range 0.30–8.37 kg). The males were significantly larger than the females in terms of both body length (ANOVA: $F_{(1,94)} = 31.29$, $p < 0.001$) and weight (Kruskal–Wallis: $H = 15.59$, $p < 0.001$).

The weight-length ratio recorded in this study (Table 1; Fig. 2) indicated isometric growth in both the female ($b = 2.91$; $t = 3.810$, $p > 0.05$) and male *E. electricus* ($b = 3.03$, $t = 114.72$, $p > 0.05$). The best fit of the growth parameters and estimates of mortality (Table 1) for the male *E. electricus* were $L_{\infty} = 186.7$ cm, $k = 0.31$ year⁻¹ and $A_{0.95} = 9.66$ years and $M = 0.7$ year⁻¹. For the females, these values were $L_{\infty} = 147.0$ cm, $k = 0.25$ year⁻¹, and $A_{0.95} = 21.4$ years and $M = 1.1$ year⁻¹.

The mean GSI values (Fig. 3) were highest in January–February (3.50 ± 1.17), followed by November–December (1.87 ± 1.73) and September–October (1.70 ± 1.31). The lowest values were recorded in March–April (0.58 ± 0.79) and May–June (0.46 ± 0.30), while July–August was intermediate (1.48 ± 1.32). These differences were significant (ANOVA: $F = 10.9017$; $p = 0.0001$) when comparing March–April and May–June with September–October (Tukey: $p < 0.05$), November–December ($p < 0.05$) and January–February ($p < 0.05$).

When analyzed by season, the GSI of the females varied significantly (ANOVA: $F = 16.9088$; $p = 0.000$), with significant differences being confirmed (Tukey: $p < 0.05$) between the high water period (mean GSI = 0.651 ± 0.52) and all other periods, that is, the flood (3.542 ± 0.59), ebb (2.045 ± 0.99), and low water (2.40 ± 1.19) periods. In the case of the males, the principal differences ($F = 4.377$, $p = 0.009$) were between the flood (0.23 ± 0.04) and the high water (0.17 ± 0.05) and ebb periods (0.18 ± 0.04), and between high water and low water (0.22 ± 0.05) (Tukey: $p < 0.05$).

The smallest mature female had a length of 73.6 cm, and the smallest male, 83.7 cm. Mean length at first gonadal maturation (L_{50}) was 85.0 cm for the

Table 1 Growth parameters for the *E. electricus* specimens from the APA do Curiaú, Amapá, Brazil

Parameter	k (year ⁻¹)	L _{max} (cm)	L _∞	W _{max} (kg)	W _∞ (kg)	A _{0.95}	M	a	b	R ²
Male	0.31	177.7	186.58	8.37	10.63	9.66	0.7	0.0014	3.03	0.91
Female	0.25	140.0	147.0	5.49	5.58	21.4	1.1	0.002	2.91	0.89

K growth constant, L_{max} maximum length, L_{∞} comprimento assintótico, W_{max} maximum weight, W_{∞} asymptotic weight, $A_{0.95}$ growth performance, M mortality, a intercept, b angular coefficient, R^2 coefficient of determination

Fig. 3 Bimstral variation in the mean Gonadosomatic Index (GSI) of the female and male *E. electricus* specimens from the APA do Curiaú, Amapá, Brazil

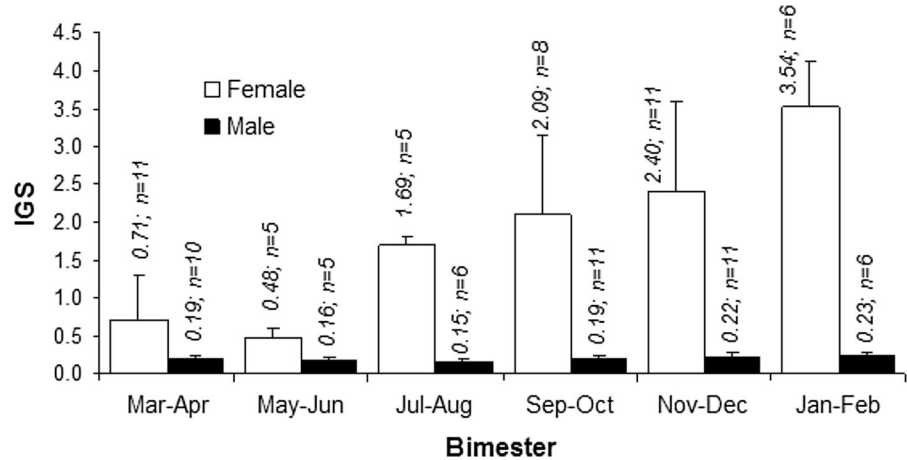
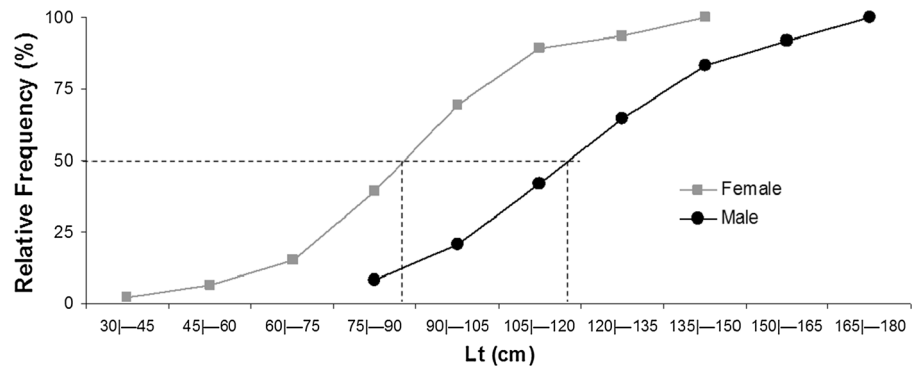


Fig. 4 Mean length at first gonadal maturation (L_{50}) in female and male *Electrophorus electricus* from APA do Curiaú, Amapá, Brazil



females, and 117.5 cm for the males (Fig. 4). The mean age at first gonadal maturation (t_{50}) was 3.45 years in the females, and 3.20 years in the males.

The ANCOVA found evidence of a significant relationship between gonadal development (GSI) and temporal variation in length in both sexes (male: $r^2 = 0.60$; $F = 3.27$; $p = 0.013$; Female: $r^2 = 0.81$; $F = 6.97$; $p = 0.0001$), whereas for weight, a significant relationship with temporal variation was found only in the females (male: $r^2 = 0.25$; $F = 1.12$; $p = 0.29$; female: $r^2 = 0.89$; $F = 26.38$; $p = 0.00001$). A significant negative correlation was

also found over the study period between the GSI and precipitation ($r_s = -0.88$; $p = 0.002$) and water levels ($r_s = -0.89$; $p = 0.01$) (Fig. 5).

Discussion

The sex ratio recorded in the present study did not deviate significantly from 1:1, and is consistent with most natural fish populations (Nikolski 1969), including other gymnotiforms, such as *Gymnotus* cf. *carapo*, *Eigenmannia virescens*, *Brachyhyppopomus draco*, and

Brachyhypopomus gauderio (Cognato and Fialho 2006; Giora and Fialho 2009; Schann et al. 2009; Giora et al. 2014). However, a considerably male-biased (3:1) sex ratio was recorded by Assunção and Schwassmann (1995) for an *E. electricus* population from Marajó Island, Brazil.

The observation of sexual dimorphism in different electric eel populations indicates that this may be a characteristic of the species, reflecting the different reproductive strategies of the two sexes, rather than a response to environmental conditions (Fairbairn et al. 2007). The males of some other Neotropical electric fish species are also larger than the females, including the apteronotids *Apteronotus leptorhynchus*, *Apteronotus bonapartii*, *Sternachogiton nattereri* and *Apteronotus lindalvae* (Dunlap et al. 1998; Hilton and Cox-Fernandes 2006; Cox-Fernandes et al. 2010; de Santana and Cox-Fernandes 2012) and the hypopomids *E. virescens*, *Eigenmannia trilineata* and *B. draco* (Kirschbaum 1979; Giora and Fialho 2009; Schann et al. 2009). However, sexual dimorphism was not found in the gymnotid *Gymnotus* aff. *carapo* (Cognato and Fialho 2006) or in hypopomid *B. gauderio* (Giora et al. 2014), while in *Apteronotus albifrons* the females are larger than the males (Dunlop et al. 1998). This indicates that sexual dimorphism in body size has arisen several times during the evolutionary history of the Neotropical electric fishes. Male body size is an indicator of the fitness of the male for reproductive activities, such as parental care (Assunção and Schwassmann 1995), and appears to be favored in some visually oriented groups

such as the cichlids and poeciliids (Ryan et al. 1990; Cacho et al. 2007; Fairbairn et al. 2007). However, the eyes (mean diameter of 4 mm; unpublished data) and visual nerves (Albert 2001) of electric eels are poorly developed, which makes female choice an unlikely driver of sexual dimorphism in these fishes.

Male body size is related positively to the frequency of the discharge of the electric organ (DOE) in gymnotiforms (Cox-Fernandes et al. 2010), and female *Brachyhypopomus pinnicaudatus*, for example, preferred larger males that had more frequent DOEs (Curtis and Stoddard 2003), indicating a possible mechanism of female selection, although more detailed research will be needed to confirm a systematic pattern, and whether in fact DOE rates increase prior to spawning, for example.

In the present study, spawning was observed at the start of the rainy season, with the variation in the gonadal development of *E. electricus* at Curiaú coinciding with the seasonal fluctuations in rainfall. Sachs and Fritsch (1881) recorded a similar pattern in *E. electricus*, and Giora et al. (2014) in other gymnotiforms, and this is in fact a typical pattern for many fish species in the lowland floodplains of the Amazon basin (Lowe-McConnell 1999; Sá-Oliveira and Chellappa 2002). However, on Marajó Island, *E. electricus* spawned during the dry season (Assunção and Schwassmann 1995). This difference may be related to the unique conditions found on this island, given that the timing of the breeding season may be influenced by local conditions (Vazzoler 1996).

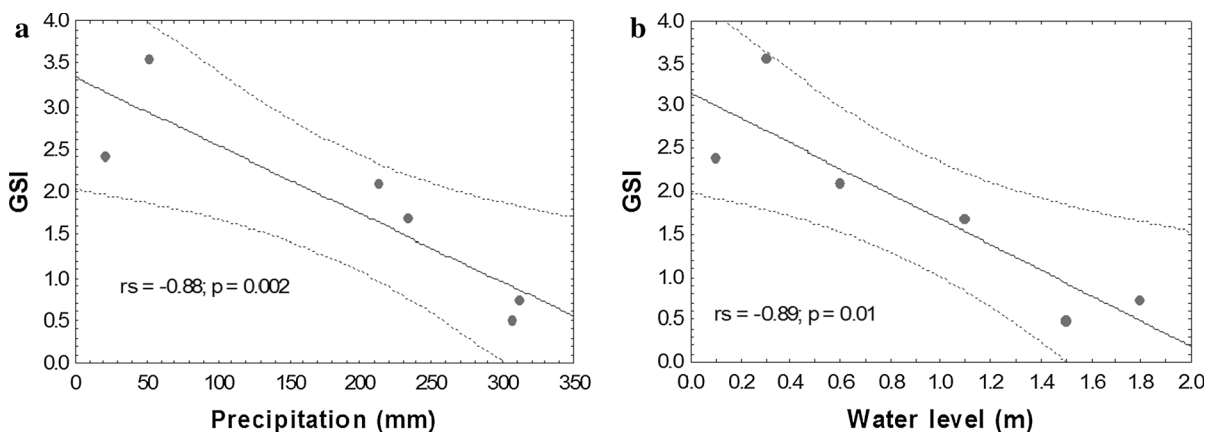


Fig. 5 Correlation between the mean GSI values and **a** precipitation and **b** water levels recorded at the APA do Curiaú, Amapá, Brazil during the study period, between March 2006 and February 2007

Sá-Oliveira and Mendes-Jr (2012) concluded that *E. electricus* spawned during the flood period in the Curiaú River due to the variation in mean fecundity and the species' spawning type. In the present study, the variation in the GSI was consistent with spawning at the start of the flood period on the floodplain of the Curiaú River. The flood and high water periods provide the best conditions for the development, growth and survival of the larvae and juveniles of most of the fish species found in this type of environment (Lowe-McConnell 1999).

The isometric growth found in both sexes of *E. electricus* may be related in part to small sample size, as well as the anatomy of species, with a relatively smaller coelomic cavity compared to other species of fish, and a comparatively large head. Elongated fishes tend to have negative allometric (Nikolsky 1969; Vazzoller 1996), as observed here in the female *E. electricus*, although a number of studies have shown that this coefficient may vary not only among species, but also between populations of the same species (Bagenal and Tesch 1978; Nahara et al. 1985).

The growth rate (k) of the male *E. electricus* was lower than that of the females, and both values were lower than those recorded for most tropical fishes (Pauly 1998; Lowe-McConnell 1999). In the study area, the principal environmental variable that may have a direct effect on the growth of *E. electricus* is the hydrological cycle and the availability of prey, which is reduced during the dry season (low water). The temperature of the water probably had little direct influence on the growth of the species, given the minor variation in this parameter at a latitude of zero degrees (Sá-Oliveira and Chellappa 2002).

The asymptotic length of *E. electricus* on the Curiaú floodplain was relatively high and was inversely related to the growth coefficient. A number of studies have shown that L_{∞} is affected by factors such as the availability of feeding resources and population density (Beverton and Holt 1957; Gomiero et al. 2007; Fonteles-Filho 2011). Mortality rates in *E. electricus* were relatively low in comparison with most other tropical fishes, which tend to have high rates of natural mortality, independently of their asymptotic size or growth rate (Pauly 1998). Two factors—predation and disease—are fundamental here, and vary according to the age of the individual, in other words, while older individuals are less susceptible to predation, they become more vulnerable

to disease through the degeneration of their tissue and loss of physiological capacities (Fonteles-Filho 1989). Growth and mortality are interactive parameters (Nikolsky 1969), given that, while growth contributes to the abundance of a species by allowing individuals to reach the reproductive phase, mortality will control this abundance.

Even if the sex ratio reduces intrasexual competition for reproductive partners (Darwin 1871), sex dimorphism in the size of electric eels suggests that there is competition for females or nesting sites (Fairbairn et al. 2007), as indicated in a second electric fish, *S. nattereri* (Cox-Fernandes et al. 2010). As electric eels males need to locate adequate sites in which to build their nests, and not all individuals can mate during the breeding season (Assunção and Schwassmann 1995), there is likely to be a limitation on the amount of suitable places for spawning. In this case, competition for spawning sites would favor the males of larger size. The monitoring the reproductive behavior of natural populations of electric eels during the maturation of the gonads in the dry season and the spawning period at the beginning of the rains could provide additional evidence of this competition. The data from this study indicate the need for the establishment of an off season from the dry season to the beginning of the rains, as well as the prohibition of harvesting individuals of less than 85 cm in length. A better understanding of spawning sites would also contribute to the development of management plans for the populations of this Amazonian species.

Acknowledgments We are grateful to the Federal University of Amapá for support, the Amapá State Environment Secretariat (SEMA/AP) for authorizing research in the APA do Curiaú, and SFF acknowledges CNPq for a research fellowship (process no. 303994/2011-8). We also thank the quilombola population of APA do Curiaú, and especially the Ramos family for their support during field expeditions.

References

- Albert JS (2001) Species diversity and phylogenetic systematics of American knifefishes (Gymnotiformes, Teleostei). *Misc Publ Mus Zool Univ Michigan* 190:1–129
- Assunção MIS, Schwassmann HO (1995) Reproduction and larval development of *Electrophorus electricus* on Marajó Island (Pará, Brazil). *Ichthyol Explor Freshw* 6(2):175–184
- Bagenal TB, Tesch FW (1978) Age and growth. In: Bagenal T (ed) *Methods for assessment of fish production in fresh*

- waters. Blackwell Scientific Publications, Oxford, pp 101–136
- Beverton RJH, Holt SJ (1957) On the dynamics of exploited fish populations. Fisheries investment series 2, vol 19. UK Ministry of Agriculture and Fisheries, London
- Cacho MSRF, Yamamoto ME, Chellappa S (2007) Mating system of the amazonian cichlid angel fish, *Pterophyllum scalare*. *Braz J Biol* 67(1):161–165
- Campos-da-Paz R (2003) Family Gymnotidae. Checklist of the freshwater fishes of South and Central America. EDI-PUCRS, Porto Alegre, pp 483–486
- Castro PMG, Cergole MC, Carneiro MH, Mucinhato CMD, Servo GJM (2002) Crescimento, mortalidade e taxa de exploração do Goete, *Cynoscion jamaicensis* (Perciformes: scianidae), na região Sudeste/Sul do Brasil. *Bol Inst Pesca* 28(2):141–153
- Chellappa S, Sa-Oliveira JC, Chellappa NT (2005) Fish fauna of a temporary lake in an Amazonian Conservation Area. *Acta Limnol Bras* 17(3):283–289
- Cognato DP, Fialho CB (2006) Reproductive biology of a population of *Gymnotus* aff. *carapo* (Teleostei: Gymnotidae) from southern Brazil. *Neotrop Ichthyol* 4:339–348
- Cox-Fernandes C, Smith GT, Podos J, Nogueira A, Inoue L, Akama A, Alves-Gomes J (2010) Hormonal and behavioral correlates of morphological variation in an Amazonian electric fish (*Sternarchogiton nattereri*: Aptereronotidae). *Horm Behav* 58(4):660–668
- Crampton WGR (1996) Gymnotiform fish: an important component of Amazonian floodplain fish communities. *J Fish Biol* 48(2):298–301
- Curtis CC, Stoddard PK (2003) Mate preference in female electric fish, *Brachyhypopomus pinnicaudatus*. *Anim Behav* 66:329–336
- Darwin CR (1871) The descent of man, and selection in relation to sex, vol 1, 1st edn. John Murray, London
- de Santana CD, Cox-Fernandes C (2012) A new species of sexually dimorphic electric knife-fish from the Amazon Basin, Brazil (Gymnotiformes: Aptereronotidae). *Copeia* 2:283–292
- de Santana CD, Vari RP, Wosiacki WB (2013) The untold story of the caudal skeleton in the electric eel (*Ostariophysi*: Gymnotiformes: *Electrophorus*). *PLoS One* 8:e68719
- Dunlap KD, Thomas P, Zakon HH (1998) Diversity of sexual dimorphism in electrocommunication signals and its androgen regulation in a genus of electric fish, *Aptereronotus*. *J Comp Physiol A* 183(1):77–86
- Fairbairn DJ, Blanckenhorn WU, Székely T (2007) Sex, size, and gender roles: evolutionary studies of sexual size dimorphism, vol 266. Oxford University Press, Oxford
- Fonteles-Filho AA (1989) Recursos Pesqueiros: Biologia e Dinâmica Populacional. Imprensa Oficial do Ceará, xv, Fortaleza
- Fonteles-Filho AA (2011) Oceanografia, Biologia e Dinâmica Populacional de Recursos Pesqueiros. Expressão Gráfica e Editora, Fortaleza
- Gayanilo FC Jr, Pauly D (eds) (1997) FAO-ICLARM stock assessment tools (FISAT). Reference manual. FAO computerized information series (Fisheries) 8. FAO, Rome, p 262
- Gayanilo FC Jr, Sparre P, Pauly D (1996) The FAO-ICLARM stock assessment tools (FISAT) user's guide. FAO Comput Inf Ser (Fisheries) 6:1–186
- Giora J, Fialho CB (2009) Reproductive biology of weakly electric fish *Eigenmannia trilineata* López and Castello, 1966 (Teleostei, Sternopygidae). *Brazil Arch Biol Technol* 52:617–628
- Giora J, Tarasconi HM, Fialho CB (2014) Reproduction and feeding of the electric fish *Brachyhypopomus gauderio* (Gymnotiformes: Hypopomidae) and the discussion of a life history pattern for gymnotiforms from high latitudes. *PLoS One* 9:e106515
- Gomiero LM, Souza UP, Braga FMS (2007) Reprodução e alimentação de *Rhamdia quelen* (Quoy and Gaimard 1824) em rios do Núcleo Santa Virgínia, Parque Estadual da Serra do Mar, vol. 7, no. 3, São Paulo, SP. Biota Neotrop
- Goulding M (1980) The fishes and the forest: explorations in Amazonian natural history. University of California Press, California
- Hilton EJ, Cox-Fernandes C (2006) Sexual dimorphism in *Aptereronotus bonapartii* (Gymnotiformes: Aptereronotidae). *Copeia* 2006(4):826–833
- Johansen K (1970) Air breathing in fishes. In: Hoar WS, Randall DJ (eds) Fish physiology, vol IV. Academic Press, London, pp 361–411
- Kirchbaum F (1979) Reproduction of the weakly electric fish *Eigenmannia virescens* (Rhamphichthyidae, Teleostei) in captivity. *Behav Ecol Sociobiol* 4:331–355
- Lowe-McConnell RH (1999) Estudos ecológicos de comunidades de peixes tropicais. Coleção Base, Edusp, São Paulo
- Mago-Leccia F (1994) Electric fishes of the continental waters of America: classification and catalogue of the electric fishes of the order Gymnotiformes (Teleostei: Ostariophysi), with descriptions of new genera and species. Fundación para el Desarrollo de las Ciencias Físicas, Matemáticas y Naturales
- Mérona BD, Rankin-de-Mérona J (2004) Food resource partitioning in a fish community of the central Amazon floodplain. *Neotrop Ichthyol* 2(2):75–84
- Nahara MY, Godinho HM, Fenerichiverani N, Romagosa E (1985) Relação Pesocomprimento e fator de condição de *Rhamdia hilarii* (Valenciennes, 1840) (Osteichthyes, Siluriformes, Pimelodidae). *Boletim Instituto Pesca, São Paulo*, 12(4): 13–22
- Nakashima S (1941) Una nueva especie de anguila eléctrica del Perú. *Boletín del Museo de Historia Natural "Javier Prado"*, Lima, 5(19): 461–465
- Nikolsky GV (1969) Fish population dynamics. Oliver and Boyd, Edinburgh
- Pauly D (1980) On the interrelationships between natural mortality, growth parameters and mean environmental temperature in 175 fish stocks. *Journal du Conseil International pour l'Exploration de la Mer* 39(3):175–192
- Pauly D (1998) Beyond our original horizons: the tropicalization of Beverton and Holt. *Rev Fish Biol Fish* 8:307–334
- Pauly D, David N (1981) ELEFAN I, a BASIC program for the objective extraction of growth parameters from length-frequency data. *Meeresforschung* 28(4):205–211
- Planquette P, Keith P, Le Bail P (1996) Atlas des poissons d'eau douce de Guyane. INRA Editions, French Guyane
- Ryan MJ, Hews DK, Wagner WE Jr (1990) Sexual selection on alleles that determine body size in the swordtail *Xiphophorus nigrensis*. *Behav Ecol Sociobiol* 26(4):231–237

- Sachs K, Fritsch G T (1881). Untersuchungen am Zitteraal: *gymnotus electricus*, Veit
- Sá-Oliveira JC, Chellappa S (2002) Fecundidade e tipo de desova do tamuatá, *Hoplosternum littorale* Hancock (Osteichthyes, Siluriformes) no Rio Curiaú, Macapá, Amapá. Rev Bras Zool 19:1053–1056
- Sá-Oliveira JC, Mendes RNG Jr (2012) Fecundidade e tipo de desova do poraquê, *Electrophorus electricus* (Lineus, 1766) (Osteichthyes: Gymnotiformes: Gymnotidae) da Área de Proteção Ambiental—APA—do Rio Curiaú, Macapá-AP. Biota Amazon 2:32–36
- Schaan AB, Giora J, Fialho CB (2009) Reproductive biology of the Neotropical electric fish *Brachyhypopomus draco* (Teleostei: Hypopomidae) from southern Brazil. Neotrop Ichthyol 7:737–744
- Sparre P, Venema SC (1998) Introduction to tropical fish stock assessment. FAO Fisheries and Technical Paper, 306/1-Rev 2: 1–407
- Taylor CC (1958) Cod growth and temperature. J Cons Int Explor Mer 23:366–370
- Vazzoler AEAM (1996) Biologia da reprodução de peixes teleosteos: teoria e prática. EDUEM/SBI, Maringá
- Wootton RJ, Evans GW, Mills LA (1978) Annual cycle in female three-spined sticklebacks (*Gasterosteus aculeatus* L.) from an upland and lowland population. J Fish Biol 12:331–343
- Zar JH (1996) Bioestatistical analysis. Prentice-Hall, New Jersey